

3. Find the area of the triangle with corners at the following three points.

$$P(1,3,1)$$

$$Q(3,0,4)$$

$$R(4,-1,2)$$

$$\vec{PQ} = \langle 2, -3, 3 \rangle \quad \vec{PR} = \langle 3, -4, 1 \rangle$$

$$\vec{PQ} \times \vec{PR} = \begin{vmatrix} 2 & -3 & 3 \\ 3 & -4 & 1 \end{vmatrix} \begin{vmatrix} 2 & -3 \\ 3 & -4 \end{vmatrix}$$

$$\text{Area} = \frac{1}{2} |\vec{PQ} \times \vec{PR}|$$

$$= \frac{1}{2} |\sqrt{9^2 + 7^2 + 1^2}|$$

$$= \frac{1}{2} \sqrt{131}$$

$$= \langle (-3 + (12)), (9 - 2), (-8 + (12)) \rangle$$

$$= \langle 9, 7, 1 \rangle$$

$$\text{area} = \underline{\underline{\frac{1}{2} \sqrt{131}}}$$